

Bhavyashree Putta

(312) 885-7833 | bhavyaputta13@gmail.com | [Linkedin](#) | [Portfolio](#) | [GitHub](#) | Chicago, IL

Professional Summary

Software Engineer with 3+ years of experience building full-stack applications, distributed systems, and data-intensive platforms that power large-scale analytical and transactional workloads. Experienced in developing responsive web applications, event-driven services, and scalable data pipelines using React, TypeScript, Java, and cloud-native technologies. Brings a systems-oriented approach to software development, with hands-on experience optimizing application performance, diagnosing production bottlenecks, and delivering reliable software from frontend interfaces to backend infrastructure.

Experience

United Way of Northern New Jersey | Software Engineer

July 2025 - Present

- Designed and developed **Java Spring Boot microservices, REST APIs, and user-facing features** powering the **United For ALICE** platform, enabling research, reporting, and data exploration across **44 U.S. states**.
- Developed **reusable React and TypeScript components** for interactive maps, data tables, and report visualizations, **reducing feature development time by 25%** through a shared component architecture.
- Engineered a **Kafka-based ingestion pipeline** that decoupled Census Bureau and state-partner data producers from downstream services through event replay, **reducing financial hardship calculation time by 45%**.
- Architected **asynchronous report generation** using **RabbitMQ task queues**, generating **40K+ report permutations** across geographic and demographic dimensions while **eliminating blocking API requests**.
- Implemented **Hibernate JPA mappings** and **optimized JPQL/SQL queries**, **reducing query latency by 35%** across **20M+ records** for interactive reporting and map-based data exploration.
- Collaborated with JPMorgan Chase engineers to build an **AI-powered conversational search platform** enabling natural-language access to 10,000+ research reports and datasets, **reducing information retrieval time by 60%**.
- Containerized Spring Boot services** with Docker and **automated builds, testing, and deployments** through **Azure DevOps pipelines**, **reducing release effort by 40%** and eliminating environment inconsistencies.
- Implemented **structured logging, distributed request tracing, health checks, and production monitoring** to accelerate root-cause analysis, **reducing incident investigation time by 30%** and improving service reliability.

University of Illinois Chicago | Full Stack Developer

Aug 2023 - May 2025

- Built **Spring Boot microservices** for **enrollment, payment, and transit-pass** provisioning serving 65,000+ students, **increasing peak registration throughput by 60%** through asynchronous processing.
- Built **Java-based RESTful service** with asynchronous workflows and schema validation, integrating queue-based processing with Azure Service Bus and automated **JUnit test suites** to **reduce integration failures by 40%**.
- Designed and optimized **relational schemas, indexing strategies, and ACID-compliant transactions**, improving query performance by 60% and enabling zero-downtime releases.
- Implemented **retry, timeout, and dead-letter recovery mechanisms** for enrollment and payment workflows to handle **Azure Service Bus outages** and downstream API failures, **reducing transaction failures by 35%**.
- Built responsive React and TypeScript interfaces for transit enrollment and payment workflows, implementing **memoization** and asynchronous data fetching to keep **First Input Delay below 50ms** during peak registration periods.
- Integrated **200+ Jest, React Testing Library, and Cypress tests** into CI/CD pipelines, **increasing code coverage by 40%**, reducing **escaped defects by 35%**, and supporting stable weekly releases.

GetLect | Full Stack Java Developer

May 2022 - Apr 2023

- Developed **Java Spring Boot REST APIs** for authentication, content delivery, and course management workflows, **reducing page load times by 30%** and improving platform responsiveness through optimized service design.
- Refactored a **monolithic application into 8 microservices** by decomposing authentication, content, and user-management modules, **reducing service dependencies and improving deployment efficiency by 40%**.
- Improved platform reliability during traffic spikes by deploying services behind **Nginx load balancers** with health checks, traffic routing, and rate limiting, maintaining **99.9% availability**.
- Migrated on-premise workloads to AWS EC2 and S3** and implemented automated backup and recovery procedures, **reducing recovery time by 50%** and improving operational resilience.
- Implemented **25+ reusable React and TypeScript components** from Figma designs, creating responsive course management and content delivery interfaces that **reduced UI revision cycles** across releases.
- Reduced database contention during peak usage by implementing **Redis caching** for frequently accessed course metadata and session data, lowering database load by **30%** and improving application responsiveness.

Projects

Distributed Replication & Consistency Simulator | C++, Python, gRPC, Raft Consensus, Fault Injection

- Built a **distributed key-value store** in C++ using the **Raft consensus algorithm**, ensuring linearizable reads and writes through leader election and **replicated state machines across a 5-node cluster**.
- Developed a **Python fault-injection framework** to simulate **network partitions**, **message delays**, and **node failures**, identifying and **resolving split-brain scenarios** during leader re-election.

DropStream | Java, Spring Boot, Kafka, Redis, PostgreSQL

- Built an **event-driven music release system** enabling real-time fan subscriptions and notifications, using **Kafka to process release events** and trigger scalable, asynchronous engagement workflows.
- Optimized event processing by **caching subscription lookups in Redis** and adding **idempotent consumer logic with retries**, preventing duplicate notifications under repeated or failed message deliveries.

Load-Aware TCP Proxy with Backpressure | C++, Linux, TCP/IP, POSIX sockets, Multithreading

- Built a **TCP proxy that routes requests** based on backend load using **connection-level metrics**, improving distribution and maintaining stable throughput under concurrent traffic.
- Implemented **backpressure via connection limits** and queue thresholds, **with timeout and retry handling** to improve stability and reduce request errors under high load.

Technical Skills

Languages: Java, Python, C++, JavaScript (ES6+), TypeScript, SQL

Backend & Distributed Systems: Spring Boot, Hibernate JPA, REST APIs, gRPC, Kafka, RabbitMQ, Microservices

Frontend: React, TypeScript, JavaScript (ES6+), HTML5, CSS3, Redux, Kendo UI, Bootstrap

Cloud & DevOps: AWS (EC2, S3, RDS), Azure, Docker, Kubernetes, Nginx, CI/CD, Git

Databases: PostgreSQL, Microsoft SQL Server, MySQL, Redis, DynamoDB, Elasticsearch

Systems & Networking: Linux, Concurrent Programming, Multithreading, TCP/IP, Inter-Process Communication (IPC)

Testing & Quality: JUnit, Mockito, Jest, React Testing Library, Cypress, Integration Testing

AI & ML: GitHub Copilot, LangChain, RAG, Vector Search, Pinecone, FAISS

Education

University of Illinois Chicago, USA

Aug 2023 – May 2025

Master of Science in Computer Science

GPA 3.88

- Featured by **Cook County Chronicle** for leading U-PASS+, expanding transit access to over 65,000 students.
- Won **Epic Cosmos Hackathon** for a data-driven HRT–breast cancer analysis using Cosmos data.